Figure 10.5. An example of soft attention where a model would learn which pixels to keep and which pixels to ignore (i.e., set to 0). Unlike the hard-attention model, the soft-attention model needs to process the entire image at once, which can be computationally demanding.



In a self-attention model (SAM), the process is quite different and more complicated. Remember, the output of a SAM is essentially a graph, except that each node is constrained to only be connected with a few other nodes (hence the “attention” aspect).

10.2.2. Relational reasoning

Before we get into the details of self-attention, let’s first sketch out how a general *relational reasoning module* ought to work. Any machine learning model is typically fed some raw data in the form of a vector or higher-order tensor, or perhaps a sequence of such tensors, as in language models. Let’s use an example from language modeling, or *natural language processing* (NLP), because it is a bit easier to grasp than processing raw images. Let’s consider the task of translating a simple sentence from English into Chinese.